5

Validity

5.1 Introduction

A test has *validity* if it measures what it purports to measure. For example, a test that is used to screen job applicants is valid if its scores are highly related to examinees' future performance on the job. A valid intelligence test discriminates among people who vary in intelligence. A valid personality test generates scores that reflect meaningful differences in personality. Both the test developer and the test user have a responsibility to ensure that the tests they provide and use are valid tests.

Validity can be assessed in several ways, depending on the test and its intended use. The three major types of validity are content validity, criterion-related validity, and construct validity. Determinations of criterion-related validity and construct validity involve the calculation and examination of correlations or other statistics. Content validity, however, does not involve any statistical calculations.

5.2 Content Validity

Content validity is established through a rational analysis of the content of a test, and its determination is based on individual, subjective judgment. There are two main types of content validity: face validity and logical validity.

Face validity, sometimes called "armchair" validity, is established when a person examines the test and concludes that it measures the relevant trait. The person making this examination can be anyone from an expert to an examinee. If people disagree, face validity is in question. Face validity may be sufficient to justify the use of some tests. The classroom exam, when carefully prepared, has face validity. For example, an arithmetic test, on the "face" of it, measures arithmetic performance. Face validity may be essential for some tests because of their intended use. For example, it would be bad for a firm's public relations if the test used to screen job applicants had no apparent relationship to the job, even if the test was effective in identifying those people most likely to be good workers. Face validity can be crucial for effective test use, although in some cases face validity is not essential if the test is valid in other ways.

Logical or *sampling validity* is a more sophisticated version of face validity. It involves the careful definition of the domain of behaviors to be measured by a test and the logical design of items to cover all the important areas of this domain. Logical validity is especially useful in the development of achievement tests. The grid in Figure 5.1 represents a possible plan for an achievement test in history. The top left element in the grid represents all the items that measure knowledge of dates in African history. Other test items would sample the content areas of the other cells in the grid. The test could be developed with either equal or unequal proportions of items representing each cell, depending on the test developer's evaluation of each cell's relative importance.

Because content validity is based on subjective judgments, the determination of this type of validity is more subject to error than are other types of validity. But, generally, establishing content validity is the first concern in the development of all tests, and items are written to meet content-validity requirements. Through statistical item-analysis techniques, the test can be revised and improved to guarantee that other aspects of good measurement are achieved. Usually, the mere fact that a test has content validity is not a sufficient justification for its use. Before it is used, the test should have proven effectiveness, such as criterion-related validity.

	African History	Asian History	European History	Latin-American History	North-American History
Dates					
Economic Changes					
Social Changes					
Political Movements					

Figure 5.1. Plan for a history-achievement test

5.3 Criterion-Related Validity

Criterion-related validity is used when test scores can be related to a criterion. The *criterion* is some behavior that the test scores are used to predict. For example, in order to have criterion-related validity, scores on a test designed to screen job applicants must be related to the criterion of work effectiveness. Similarly, in order to have criterion-related validity, scores on a school-admissions test must be related to some relevant criterion, such as later grade-point averages of the accepted students or percentage of students who complete their programs and receive degrees.

Criterion-related validity typically is expressed as a correlation coefficient —the correlation between the test (*predictor*) score and the criterion score. This correlation is symbolized ρ_{XY}, where X is the test score and Y is the criterion score. The *validity coefficient,* ρ_{XY}, is estimated in one of two ways, resulting in either a predictive- or a concurrent-validity estimate.

Predictive validity involves using test scores to predict future behavior. A *predictive-validity coefficient* is obtained by giving the test to all relevant people, waiting a reasonable amount of time, collecting criterion scores, and calculating the validity coefficient. For example, the predictive validity for an employment test would be established by testing every job applicant, hiring every applicant, waiting a few weeks or months until the criterion can be assessed reasonably and reliably (for example, by supervisors' ratings or by some other measure of job performance), and correlating the scores on the predictor (the test) and the criterion (job performance). This procedure would give a good indication of how well the test scores predict future behavior, but it can be expensive and time consuming. If the employer is not willing to hire all applicants, there probably will be restriction of range on both the predictor and criterion. As shown in Section 2.10, restriction of range generally attenuates (reduces) an observed correlation coefficient, leading to an underestimate of the predictive-validity coefficient.

When a test is used to predict future behavior, predictive validity should be established. A less desirable alternative in this situation is to use a concurrent-validity coefficient. A *concurrent-validity coefficient* is a correlation between test and criterion scores when both measurements are obtained at the same time. For the employment-test examples just given, the concurrent-validity coefficient would be the correlation between the predictor and the criterion scores when both measurements are made on present employees at the same time, rather than being separated by a time interval. This often involves a large restriction of range, especially on the criterion, since most people who could not or do not perform satisfactorily on the job would not be hired or would be fired by the time the validity study took place. Thus, the concurrent-validity coefficient would tend to underestimate a predictive-validity coefficient.

Concurrent-validity coefficients are appropriate when the test scores are used to estimate a concurrent criterion rather than to predict a future criterion. For

example, the *Minnesota Multiphasic Personality Inventory* (MMPI) is an extremely long (almost 600-item) personality test. A counselor might give a brief personality test to all clients and reserve the MMPI for clients whose scores on the brief test suggest that they might have particularly interesting MMPI scores. This two-step process would be reasonable only if the scores on the brief test were related to the MMPI scores. The counselor needs to demonstrate concurrent validity for the brief test as the predictor and the MMPI scores as the criteria.

5.4 Correction for Attenuation

We pointed out in Section 4.1 that the reliability of a test affects the validity of the test, since, theoretically, a test cannot correlate more highly with any other score than it correlates with its own true score; that is, $\rho_{XY} \leq \rho_{XT}$ or $\rho_{XY} \leq \sqrt{\rho_{XX'}}$. If both a test score, X, and a criterion score, Y, are unreliable, the validity coefficient, ρ_{XY}, may be reduced in value (attenuated) relative to the value of the validity coefficient that would be obtained if X and Y did not contain measurement error. An estimate of the correlation that would be found if X and Y were perfectly reliable can be obtained with the *correction for attenuation,* which was first developed by Spearman (1904). The correction for attenuation gives the correlation between T_X and T_Y under the classical model,

$$\rho_{T_X T_Y} = \frac{\rho_{XY}}{\sqrt{\rho_{XX'}\rho_{YY'}}}. \tag{5.1}$$

In this equation, $\rho_{T_X T_Y}$ is the correlation between the true score for X and the true score for Y or the correlation between X and Y if there is no error of measurement in either observed score; ρ_{XY} is the correlation of observed scores X and Y, which includes error of measurement; and $\rho_{XX'}$ and $\rho_{YY'}$ are the reliabilities of X and Y. The correction for attenuation appears as Conclusion 14 in Chapter 3.

We can also correct for attenuation due to unreliability only in the predictor or only in the criterion. For example,

$$\rho_{XT_Y} = \frac{\rho_{XY}}{\sqrt{\rho_{YY'}}} \tag{5.2}$$

would be the correlation between X and Y if the measurement of Y were perfectly reliable. This would be a reasonable correction when a validity coefficient is attenuated due to unreliability in the criterion measurement.

The correction for attenuation has one application that is particularly useful. Suppose that a single test is needed to predict an important criterion. There are several brief tests (of approximately the same length) under consideration, each with a different reliability and validity. The test developer wants to choose the most promising test and lengthen it to improve its reliability. Given the data in Table 5.1, which test is best? Tests A and B have the same validity coefficients but different reliabilities. Test C has a lower validity coefficient but is also the least reliable.